



# Python Basics

## Python comment

```
In [22... # single line comment

# shipt+alt+A

# multi line comment (doc string) => """and""" or '''manju'''

"""
The is a
multiple
line
comments
Example
"""
```

```
Out[22]: ' \nThe is a \nmultiple \nline \ncomments\nExample\n\n'
```

## python Keywords

```
In [23... # python case sensitive PROGRAMMING LANGUAGE
# True, False, None => is capital
```

## Data Types

- Integer – 1, 1244, -1234
- Float – 1.0, 1e10, 123.324
- complex – 1+0j (j represents that its complex)
- string – “Hello world”, “a”, ‘a’
- boolean – True, False

## Python Variable

Rules:

- Starts with Alphabets or underscore ( **a-z, A-Z, \_** )
- Then Can be a combination of Alphabets, Underscore & Number
- Cannot be Keywords

```
In [24... # Integer – 1, 1244, -1234
```

```

name = "Jeevan"
age = 15

manju_nathan = "Python"

a = 100
A = 200
a = 400

_and = "Kumar"

A10 = 1000

_____ = "Honey"

# B% = 40.55

# a,b,c,d= "Hai","Hello","Welcome"

a=b=c=200

b=c
a=b
print(c)
print(b)
print(a)

```

```

200
200
200

```

## Python Data Types

- Integer – 1, 1244, -1234
- Float – 1.0, 1e10, 123.324
- complex – 1+0j (j represents that its complex)
- string – “Hello world”, “a”, ‘a’
- boolean – True, False

```

In [25... # Integer – 1, 1244, -1234

a = 500
a1 = -500

# Float – 1.0, 1e10, 123.324

b = 50.55
b1 = -50.55

b2 = 2e7

```

```
# complex - 1+0j          (j represents that its complex)

c = 20j

# string - "Hello world", "a", 'a'

d = "Hai"
d1 = "100" + "50"

# boolean - True, False

e = True
e1 = False
print(e,type(e))
print(e1,type(e1))
```

```
True <class 'bool'>
False <class 'bool'>
```

## print() - Standard Functions

```
In [26... print(100+50-20)
print("Hello : Kumar")
```

```
130
Hello : Kumar
```

## Escape sequences

- \ - for newline in editor
- \\ - prints \
- \' - prints '
- \" - prints "
- \n - prints a new line
- \t - prints a tab

```
In [27... print("hello \\world\\")
print("10\\100")
print('I\'am Python world')
print("Hello\\\"World\\")
print("Hello\\nWorld") # \n=> New Line
print("Hello\\tWorld") # \t => tab
```

```
hello \\world\\
10\\100
I'am Python world
Hello"World"
Hello
World
Hello      World
```

```
In [28... print("Lorem Ipsum is simply dummy text of the printing and typesetting i
        Lorem Ipsum has been the industry's standard dummy text ever since
        when designers at Letraset and James Mosley, the librarian at St Br
        Printing Library in London, took a 1914 Cicero translation and scr
        it to make dummy text for Letraset's Body Type sheets. It has survi
        only many decades,but also the leap into electronic typesetting, re
        essentially unchanged. It was popularised thanks to these sheets an
        recently with desktop publishing software like Aldus PageMaker and
        Word including versions of Lorem Ipsum.")
```

Lorem Ipsum is simply dummy text of the printing and typesetting industr  
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ishing software like Aldus PageMaker and Microsoft Word including v  
ersions of Lorem Ipsum.

```
In [29... print("""\tLorem Ipsum is simply dummy text of the printing and typesetti
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```

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publishing software like Aldus PageMaker and Microsoft Word including ver  
sions of Lorem Ipsum.

```
In [30... # end, sep

# sep

print(1,2,3,4,5,sep="**")
print(6,7,8,9,10)

1**2**3**4**5
6 7 8 9 10
```

```
In [31... # end
```

```
print(1,2,3,4,5,end=" ")

print(6,7,8,9,10,end=" ")
print(11,12,13,14,15)
```

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

## Trio : value, id, type

```
In [32]: a = 100
print(a)
print(type(a))
print(id(a))
```

100  
<class 'int'>  
140721670276632

## Type casting

```
In [ ]: a = 500

a1 = float(a)
a2 = str(a)
a3 = bool(a)

b = 40.55
b1 = int(b)
b2 = str(b)
b3 = bool(b)

c = "hello"
# c1 = int(c)
# c2 = float(c)
c3 = bool(c)

d = "500"
d1 = int(d)
d2 = float(d)
d3 = bool(d)

e = bool("") # => bool => False
e1 = bool(0) x
print(e,type(e))
print(e1,type(e1))
```

False <class 'bool'>  
False <class 'bool'>

## input() - function

```
In [57... num = 100
print("The value of num = ",num)
print("The value of num = ",num)
print("The value of num = ",num)
print("The value of num = ",num)
```

```
The value of num = 100
The value of num = 100
The value of num = 100
The value of num = 100
```

```
In [60... print("Enter a Number :")
num = input()
print("The value of num = ",num)
```

```
Enter a Number :
The value of num = 500
```

```
In [61... num = input("Enter a Number :")
print("The value of num = ",num)
```

```
The value of num = 500
```

```
In [63... num1 = input("Enter a Number Num1 :")
print("The value of num1 = ",num1)
```

```
num2 = input("Enter a Number Num2:")
print("The value of num2 = ",num2)
```

```
Total = num1+num2
print("The sum of :",Total)
```

```
The value of num1 = 200
The value of num2 = 300
The sum of : 200300
```

```
In [70... num1 = input("Enter a Number Num1 :")
x = int(num1)
print("The value of num1 = ",x,type(x))
```

```
num2 = float(input("Enter a Number Num2:"))
print("The value of num2 = ",num2,type(num2))
```

```
Total = x+num2
print("The sum of :",Total)
```

```
The value of num1 = 50 <class 'int'>
The value of num2 = 60.0 <class 'float'>
The sum of : 110.0
```

## del keyword

```
In [68... num1 = input("Enter a Number Num1 :")

x = int(num1)
del num1
print("The value of X = ",x,type(x))
print("The value of num1 = ",num1,type(num1))
```

The value of X = 800 <class 'int'>

```
-----
NameError                                Traceback (most recent call last)
Cell In[68], line 6
      4 del num1
      5 print("The value of X = ",x,type(x))
----> 6 print("The value of num1 = ",num1,type(num1))

NameError: name 'num1' is not defined
```